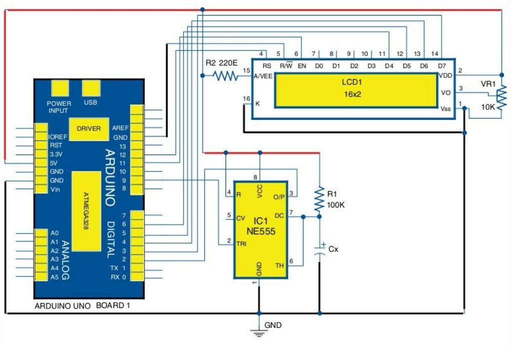

Fig. 5: Circuit diagram of Arduino based digital capacitance meter with NE555 timer in monostable mode

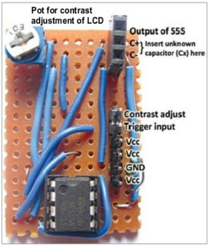

Fig. 6: NE555 timer connection in authors' breakout board

ino) uploaded to Arduino uses an interrupt-within-an-interrupt handler, that is, a two-level nested interrupt sub-routine (ISR).

The first interrupt-handler gets executed whenever the timer output makes low-to-high transition, and the second ISR is called from within the first when timer output makes high-to-low transition. Thus, time for which the output of the timer remains high is obtained by calculating the time difference between two such consecutive interrupts, which is equal to T.

Thus, value of Cx is given as:

Cx = T/(1.1 × R1)

Measured value of Cx (in µF) is then displayed on a 16 × 2-character LCD and serial monitor of Arduino IDE.

NE555 timer. NE555 timer IC1 operates in monostable multivibrator mode, where time for the output goes high, after applying high-low-high pulse from pin 9 of Arduino, which is controlled by R1 and Cx connected externally to the IC.

Output of timer pin 3 is connected to interrupt pin (pin 2) of Arduino. Pins 4 and 8 of the IC1 are connected to + 5V connector of Board 1. R1 of 100kΩ is connected between pins 8 and 7. IC1 shares the same ground with Arduino board. The authors' designed breakout board for the timer is shown in Fig. 6.

Software

The code (capacitance2.ino) written in Arduino programming language uses LiquidCrystal.h header file provided by Arduino library for working with the LCD.

attachInterrupt(0,analyze1,RISING) function calls the interrupt handler named analyze1 whenever output of IC1 connected to interrupt 0 pin (pin 2) of Arduino makes low-to-high, that is, rising-edge transition.

attachInterrupt(0,analyze2,FALLING) function calls the interrupt handler named analyze2 whenever output of IC1 connected to pin 2 of Arduino makes high-to-low, that is, falling-edge transition.

The high-low-high trigger pulse applied to pin 2 of IC1 is generated by pin 9 of Arduino using the following code within void loop() function. Refer source code for the same.

```
void loop ( ){
digitalWrite(9, HIGH);
delay(10);
digitalWrite(9, LOW);
delay(1);
digitalWrite(9, HIGH);
while (1);
}
```

Note. To test a new capacitor (Cx), connect the capacitor and press Reset on Arduino Uno board. EFY

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EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download at *source.efymag.com*